



Cycle-Aware Adaptive Horizon Model Predictive Control for Vehicle Trajectory Optimization at Signalized Intersections

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Abstract: The increasing complexity of modern urban traffic networks demands intelligent control strategies that can anticipate and adapt to dynamic traffic conditions. Model Predictive Control (MPC) is a framework that optimizes vehicle control by predicting future states and respecting real-time constraints, such as traffic signals at intersections. However, the computational complexity of MPC increases significantly with the number of decision variables and constraints, which is directly proportional to the length of the prediction horizon, creating a critical trade-off between control performance and computational efficiency. To address this challenge, this paper proposes an adaptive-horizon optimal driving (AHOD) bi-level optimization framework that incorporates a novel time-step discretization for real-time trajectory optimization and integrates it into a full traffic signal cycle. Unlike conventional MPC, which employs uniform time discretization leading to exponential growth in decision variables with horizon length, the proposed AHOD framework assigns finer time steps near signal phase transitions and coarser steps in the distant horizon, maintaining a fixed number of optimization nodes regardless of cycle length. The proposed framework comprises two controllers: the upper and lower controllers. The Upper controller employs finer resolution at critical times of signal change and coarser resolution in distant horizons, thereby reducing computational cost while maintaining prediction accuracy. The lower controller applies a practical MPC scheme to generate realtime control actions that are consistent with the long-term constraints of the upper controller. Simulation results demonstrate that the proposed framework achieves up to 17.6% fuel savings compared to traditional human driving and reduces computation time by approximately 61% compared to long-horizon MPC, while maintaining comparable control performance. The proposed framework enables real-time, cycle-aware predictive control for connected and automated vehicles (CAVs), and establishes a practical basis for embedding long-horizon prediction within an MPC-based trajectory-planning framework.

Keywords: Adaptive-horizon prediction; Adaptive discretization; Trajectory optimization; Intelligent traffic control; Computational efficiency

1 Introduction

Urban traffic congestion at signalized intersections remains a critical challenge in modern transportation systems, despite extensive developments in intelligent transportation systems (ITS) [1]. According to a recent report [2], U.S. drivers alone wasted approximately 2.8 billion gallons of fuel and 9.8 billion hours in delays annually, equivalent to nearly 63 hours per commuter in 2024. The growing traffic demand on urban networks often exceeds the capacity of existing infrastructure, leading to prolonged idling, stop-and-go flow, and additional 14% greenhouse gas (GHG) emissions compared to steady-flow conditions [3–5]. To address these challenges, predictive control strategies based on vehicle-to-infrastructure (V2I) communication have been proposed to enhance driving performance, safety, and fuel efficiency.

The development of connected and automated vehicles (CAVs) has generated research interest in intersection optimization and predictive traffic management [6]. CAVs can exchange real-time information on signal phases, queue

lengths, and neighboring vehicle states using vehicle-to-vehicle (V2V) and V2I communication [7, 8]. Consequently, this data can be used by predictive control strategies to anticipate dynamic traffic conditions. Among existing optimization-based control strategies, Model Predictive Control (MPC) is recognized as a leading framework for vehicle trajectory optimization due to its ability to handle real-time constraints and anticipate future system states [9]. Several studies have demonstrated the benefits of MPC at intersection coordination. For example, Sun et al. [10] and Pourmehrab et al. [11] reported about 15–25% fuel savings using MPC-based eco-driving, while Chen et al. [12] showed that hierarchical MPC improves throughput in congested intersections. Further research has extended these approaches to platoon coordination [13] and advisory eco-driving [14]. Despite these advantages, current MPC implementations typically use short prediction horizons, which effectively address near-term inefficiencies.

Extending the prediction horizon enables MPC to cover the entire traffic cycle. However, each additional step introduces new decision variables, constraints, and dynamic equations, leading to near-linear growth in problem size and often exponential increases in solver time [15, 16]. This fundamental trade-off between predictive capability and computational complexity remains an important limitation in long-horizon MPC design. Moreover, the discretization of continuous-time into a finite number of optimization steps introduces additional complexity. Finer timestep sizes improve numerical accuracy but significantly decrease the overall prediction horizon. Conversely, coarser discretization covers longer time intervals at the cost of prediction accuracy and constraint satisfaction. Therefore, achieving an effective balance between numerical resolution and computational efficiency is crucial for real-time MPC implementation in traffic applications. Numerous strategies have been proposed to mitigate the computational burden of MPC. Adaptive-horizon Model Predictive Control (AHMPC) methods dynamically adjust the prediction horizon based on real-time system requirements [15, 17–19]. Despite these advances, existing adaptive-horizon MPC methods still exhibit notable shortcomings in the context of signalized intersection control. First, while adaptive-horizon approaches dynamically adjust prediction length, AHMPC often introduces additional computational overhead at each time step to determine optimal horizon length. Second, variable discretization or move-blocking [20, 21] and pseudospectral methods [22, 23] improve numerical accuracy and reduce problem dimension. However, these approaches are limited by constraint violation in coarser prediction regions and low numerical accuracy during signal transitions. Third, Koopman operator theory has recently been used to reformulate nonlinear MPC into lifted linear predictors, which significantly reduces computational complexity [24, 25]. However, this approach requires extensive offline data collection across multiple routes, speed limits, road grades, and stopping scenarios, followed by training and validation of the learned linear system. Additionally, the generalization of data-driven MPC to unobserved traffic patterns remains uncertain.

To address the above limitations, this study proposes an adaptive-horizon optimal driving (AHOD) bi-level optimization framework that extends the prediction horizon to cover entire signal cycles while maintaining computational feasibility. The proposed AHOD framework addresses limitations by fixing the number of optimization nodes while allowing non-uniform time steps, maintaining constant problem dimensionality regardless of signal cycle length, and separating long-horizon strategic planning from short-horizon real-time execution. The main contributions of this work are summarized below:

- To enable vehicles to anticipate traffic signal cycles, an AHOD framework is developed, in which a long-horizon predictive optimization problem is formulated to capture upcoming signal phase transitions and downstream interactions.
- To extend the effective prediction horizon without increasing the number of optimization variables, an adaptive horizon strategy is introduced at the upper controller, allowing fine temporal resolution near critical events (e.g., signal phase changes) and coarser resolution in distant horizons.
- To ensure real-time feasibility and safe vehicle operation, a lower-level MPC controller is designed to track the long-horizon reference trajectory while ensuring safe car-following and collision-avoidance under real-time execution limits.
- Through extensive numerical simulations under varying signal timings and traffic conditions, the proposed AHOD framework is shown to improve fuel efficiency, average speed, and travel time, while significantly reducing computational burden compared to conventional long-horizon MPC.

The remainder of the paper is structured as follows. Section II describes the methodology of the proposed study. Section III presents the results and discussion, analyzing computational complexity and control performance. Section IV concludes the paper with design guidelines and directions for future research.

2 Methodology

Vehicles approaching an intersection must make timely decisions depending on the conditions of the traffic signal and downstream traffic. Accordingly, the host vehicle’s longitudinal behavior is characterized through its state variables, position x_h and velocity v_h . The control input a_h is determined based on both the host vehicle’s current state and the state of the preceding vehicle, represented by its position x_p and velocity v_p , as depicted in Figure 1. The objective of the proposed framework is to determine the optimal acceleration trajectory that covers the total

cycle length and minimizes computational burden.

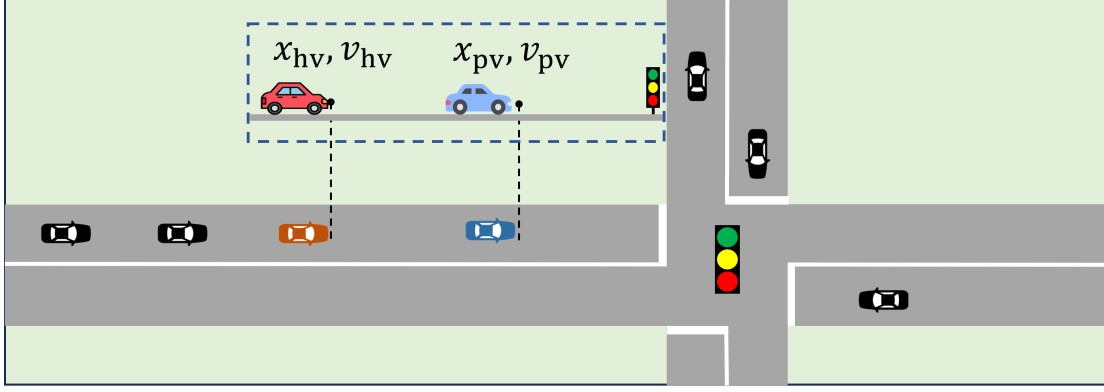


Figure 1. A schematic illustration of a vehicle approaching the signalized intersection

2.1 Vehicle Dynamics and Traditional Model Predictive Control

The longitudinal motion of a vehicle k can be expressed using a fundamental continuous-time kinematic relationships formulation, which has been widely used in longitudinal motion control [26, 27], as:

$$x_k(t+1) = x_k(t) + v_k(t)\Delta t + 0.5a_k(t)\Delta t^2, \quad (1)$$

$$v_k(t+1) = v_k(t) + a_k(t)\Delta t, \quad (2)$$

where, x_k , v_k , and a_k denote position, velocity, and acceleration at time t with time step Δt . During a red signal, the acceleration is determined by considering the more restrictive constraint between the intersection stop-line position X_{int} and the preceding vehicle position, and is defined as:

$$a_k(t) = \min\{f(d_{k-1}(t), v_k(t), v_{k-1}(t)), f(d_{\text{red}}(t), v_k(t), 0)\}, \quad (3)$$

where, d_{k-1} presents the distance to the preceding vehicle $k-1$, and d_{red} is the desired stopping position corresponding to the intersection stop line.

In this study, the Intelligent Driver Model (IDM) developed by Treiber et al. [28] is used as a benchmark traditional human driving model (THD). The instantaneous acceleration $a_h(t)$ at time t can be expressed as a function of the relative distance and velocities as:

$$f_{\text{thd}}(\Delta x_r, v_h, v_p) = a \left(1 - \left(\frac{v_h}{V_d} \right)^4 - \left(\frac{D^*(v_h, v_p)}{\Delta x_r} \right)^2 \right), \quad (4)$$

where, $\Delta x_r = x_p - x_h$ is distance between vehicles or distance to intersection stop-line position, V_d is desired velocity, and $D^*(v_h, v_p)$ is the desired gap from the preceding vehicle obtained as:

$$D^*(v_h, v_p) = R_0 + v_h t_{\text{hd}} + \frac{v_h(v_h - v_p)}{2\sqrt{ab}}, \quad (5)$$

where, R_0 , t_{hd} , a , and b represent the minimum distance, the safe time gap, the maximum acceleration, and the comfortable deceleration rates, which naturally vary depending on driving style. The resulting acceleration is then applied as the control input $a_k(t)$ to the longitudinal motion model described by Eqs. (1)–(2).

The IDM ensures collision-free vehicle control in urban traffic. However, despite its computational advantages, IDM focuses solely on vehicle states at a present time, neglecting future states, which often results in suboptimal traffic flow, particularly at signalized intersections. Specifically, the braking behavior of traditional human car-following models at red signals is not energy-efficient. Additionally, the lack of compactness in vehicle movement when transitioning from idling to quick starts at red-to-green signal changes can negatively impact energy consumption and maximum intersection throughput. On the other hand, the prediction horizon T_h of conventional MPC is discretized into uniform intervals of fixed length Δt . The optimization problem is solved repeatedly at each control step, producing an optimal control sequence that minimizes a cost function while satisfying dynamic and safety constraints. However, as T_h increases to cover longer traffic signal cycles, the number of decision variables increases linearly with $\frac{T_h}{\Delta t}$, resulting in a significant computational load.

2.2 Adaptive Horizon Optimal Driving

To address this limitation, we propose an AHOD bi-level optimization framework to improve intersection traffic flow performance and computational load. Specifically, the framework consists of two in-vehicle controllers, namely an upper controller for trajectory prediction and a lower controller for real-time execution, as shown in Figure 2. The upper controller introduces an adaptive horizon for trajectory prediction, in which the time intervals within the prediction horizon are non-uniform. Shorter steps are used for the initial step and signal phase boundaries to capture transient dynamics accurately, while longer steps are applied for distant predictions where system behavior changes more slowly. On the other hand, the lower controller integrates the generated trajectory, ensuring compliance with safe car-following constraints.

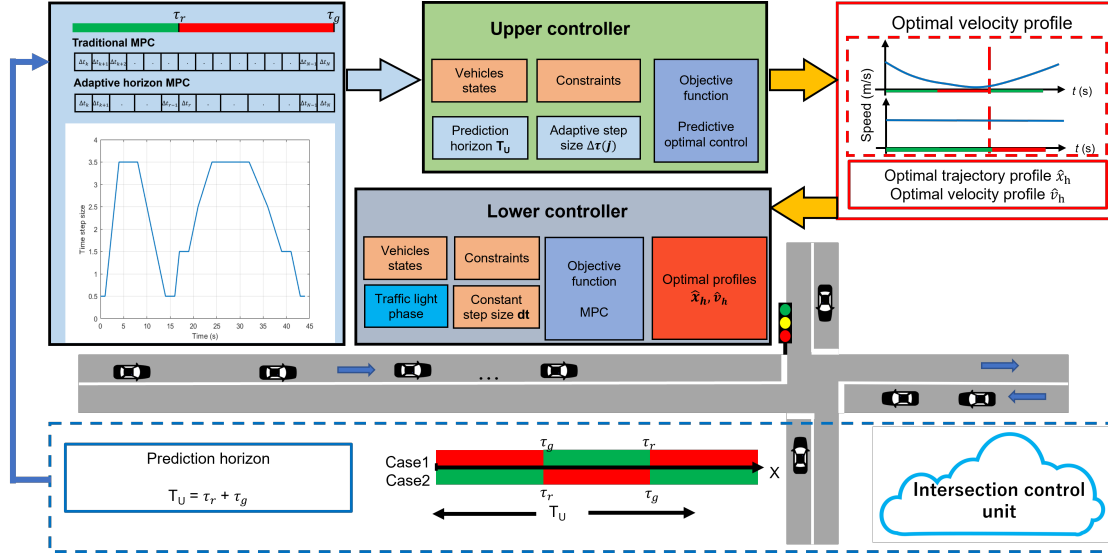


Figure 2. Illustration of interaction between the upper and lower controllers, the intersection control unit, and autonomous vehicles. The system processes real-time data to generate an optimal velocity trajectory for vehicles approaching the signalized intersection

2.2.1 Upper controller

The upper controller represents long-horizon trajectories efficiently without increasing the problem size in proportion. The adaptive discretization $\Delta\tau(j)$ is introduced to capture transient dynamics near signal phase transitions while limiting the number of decision variables for long-horizon prediction. Specifically, a fixed number of prediction nodes N is maintained, while non-uniform time-step lengths are assigned across the horizon. The signal phase switching times τ_g and τ_r are included as grid points to ensure alignment with traffic signal constraints. Finer discretization is applied in the initial step and around phase boundaries, whereas coarser steps are used in the distant horizon where vehicle dynamics evolve more gradually, as described in Algorithm 1.

When the vehicle enters the region where signal phase information is available, the upper controller determines whether the intersection can be cleared within the current green phase. In cases when a vehicle cannot cross an intersection during the remaining green signal, an optimal deceleration strategy is generated to avoid unnecessary stopping.

Specifically, for a vehicle approaching a signalized intersection and given the upcoming signal phase switching times τ_g and τ_r , the following predictive optimization problem is formulated at the upper-controller:

$$\min_{a_h(\cdot)} J_U(x_h(0), v_h(0), a_h(\cdot)) \quad (6)$$

with a prediction horizon of N steps,

$$J_U(\cdot) = \sum_{j=0}^{N-1} \omega_1(j) a_h^2(j) + \sum_{j=1}^N \omega_2(j) (V_d - v_h(j))^2, \quad (7)$$

subject to:

$$\begin{cases} a_{\min} \leq a_h(j) \leq a_{\max}, \\ 0 \leq v_h(j) \leq v_{\max}, \\ x_h(\tau_g) \leq X_{\text{int}}, \end{cases} \quad (8)$$

where, the weight coefficients $\omega_1(j)$ and $\omega_2(j)$ are defined as functions of the discretization step j , which corresponds to varying time-step lengths across the prediction horizon. The decision variable of the upper controller is the long-horizon acceleration sequence $\{a_h(j)\}_{j=0}^{N-1}$ defined over a nonuniform temporal grid $\Delta\tau(j)$. Given the initial state $(x_h(0), v_h(0))$, the optimal acceleration sequence is propagated through the longitudinal vehicle dynamics defined in Eqs. (1)–(2) to generate the reference position and velocity trajectories $\{\hat{x}_h(j), \hat{v}_h(j)\}_{j=1}^N$. Since the lower controller operates on a uniform time grid with fixed step Δt , the reference trajectories are linearly interpolated from the nonuniform upper grid onto the lower controller's uniform grid. The upper controller is triggered once when the vehicle enters the zone, where signal information available, and the interpolated reference trajectories $\{\hat{x}_h(t), \hat{v}_h(t)\}$ are computed and transmitted to the lower controller as interface signals, where it can be at every control step to ensure real-time feasibility and safety through safe car-following and collision-avoidance constraints. The upper controller can be retrIGGERED periodically to update the reference profile in response to changes in traffic conditions. However, in this work, the reference is computed once per vehicle entry for simplicity and remains fixed until the vehicle clears the intersection.

Algorithm 1 Adaptive horizon for upper controller

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1: Initialize current time  $\tau_0 = 0$ 
2: Given signal switching times  $\tau_g, \tau_r$ 
3: Minimum and maximum step sizes  $\Delta\tau_{\min}, \Delta\tau_{\max}$ 
4: Total number of nodes  $N$ 
5: Initialize grid  $\mathcal{T} = \{\Delta\tau_{\min}\}$ 
6: for  $j = 1$  to  $N$  do
7:   if  $\tau_{j-1}$  is close to  $\tau_g$  or  $\tau_r$  then
8:      $\Delta\tau(j) = \Delta\tau_{\min}$ 
9:      $\tau_j = \Delta\tau(j)$ 
10:  else
11:     $\Delta\tau(j) = \min(\Delta\tau_{\max}, \Delta\tau(j-1) + \delta)$ 
12:     $\tau_j = \tau_{j-1} + \Delta\tau(j)$ 
13:  end if
14:  Append  $\tau_j$  to  $\mathcal{T}$ 
15: end for
16: return  $\mathcal{T}$ 

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2.2.2 Lower controller

The lower controller operates independently, with a shorter horizon, to support safe car-following behavior. Furthermore, when the upper controller provides reference profiles, the lower controller incorporates them into the optimization problem as additional constraints. In cases of conflict between reference tracking and safety, the hard safety constraints take strict priority, and the reference trajectory is treated as a soft target through the separation error penalty term. Correspondingly, at each time step for a prediction horizon size of T , the following optimization problem is solved as:

$$\begin{aligned}
& \min_{a_h(\cdot)} J_{\text{AHOD}}(x_h(t), v_h(t), a_h(\cdot)) \\
& = \sum_{t=0}^{T-1} \omega_a a_h^2(t) + \sum_{t=1}^T \omega_v (V_d - v_h(t))^2 + \omega_{\text{hw}}(t) X_{\text{err}}^2(t),
\end{aligned} \tag{9}$$

subject to:

$$\begin{cases}
a_{\min} \leq a_h(t) \leq a_{\max}, \\
0 \leq v_h(t) \leq v_{\max}, \\
x_h(t) \leq \hat{x}_p(t) - t_{\text{hw.min}} v_h(t) - R_0, \\
v_h(T) \leq \hat{v}_h(T), \text{ for } T\Delta t \leq \tau_g, \\
x_h(T) \leq \hat{x}_{\text{int}}(T), \text{ for } T\Delta t \leq \tau_g,
\end{cases} \tag{10}$$

where, R_0 is the minimum inter-vehicle distance, w_a and w_v are positive acceleration and velocity weights, $X_{\text{err}}(t) = t_{\text{hw}} v_h + x_h - x_p + R_0$ is the separation error, and $\omega_{\text{hw}}(t)$ is the weight factor of safe headway as a function of time and can be expressed as:

$$\omega_{\text{hw}}(t) = a_1 e^{-a_2 \tanh(a_3(t_h(t) - t_{\text{hw}}))}, \tag{11}$$

where, a_1 , a_2 , and a_3 are constant coefficients and $t_h(t) = (x_p - x_h - R_0)/(v_h + \alpha)$ is the instantaneous time headway between vehicles with small constant α to avoid singularity. The coefficient a_1 governs the maximum

penalty magnitude applied when the instantaneous time headway falls critically below the safe headway t_{hw} . The coefficient a_2 controls the sharpness of the penalty transition around the safe headway boundary, while a_3 determines the sensitivity of the weight to deviations in instantaneous time headway $t_h(t)$ from the desired safe headway t_{hw} .

3 Results and Discussion

To demonstrate the effectiveness of our proposed method, we used a traffic simulator developed in MATLAB [29] to model a single-lane urban signalized intersection located 300 m downstream of the initial vehicle positions. The safe time gap and minimum distance between vehicles for the THD are set to $t_{hd} = 1.3$ s and $R_0 = 2$ m. Other simulation parameters are given as $a = 1.5$ m/s², $b = 2.5$ m/s², $V_d = 50$ km/h and $L = 4$ m. The control step is discretized at $\Delta t = 0.5$ s. For the AHOD, different prediction horizon lengths $N = \{10, 20, 30, 40\}$ are examined. The upper controller is configured with $N = 20$ nodes, where the minimum time step is set to 0.5 s, and the maximum time step $\Delta\tau_{max}$ is set such that the total prediction horizon covers the full signal cycle length ensuring that both signal switching times τ_g and τ_r are always captured as grid points within the adaptive horizon. All optimization problems are solved using the Sequential Quadratic Programming (SQP) algorithm implemented in MATLAB's fmincon solver. Analytical gradients of the objective function are provided to improve convergence efficiency. The maximum number of function evaluations is set to 5000, with optimality tolerance 10^{-6} and constraint tolerance 10^{-4} . The constraint tolerance is relaxed to ensure numerical feasibility. Warm-starting is enabled for all MPC-based controllers by initializing each optimization with the shifted solution from the previous time step. Identical solver settings are used across all horizon lengths to ensure fair computational comparison.

The velocity deviation and acceleration weight factors are tuned as $\omega_a = 15$ and $\omega_v = 0.4$, respectively. The prediction horizon for the upper controller is divided into 20 uneven steps $\Delta\tau(j)$, while the lower controller has a constant time step of 0.5 s. To maintain consistency with the uneven time-steps, the weighting coefficients of the acceleration and velocity terms of the proposed system are defined as $\omega_1(j) = \omega_a \Delta\tau(j)/\Delta t$ and $\omega_2(j) = \omega_v \Delta\tau(j)/\Delta t$, respectively. Constant coefficients of separation error are $a_1 = 0.03$, $a_2 = 8.88$, $a_3 = 0.27$, and $t_{hw} = 1.3$ s. Initially, two traffic signal scenarios are considered to evaluate the performance of the AHOD for a single car under different signal phase durations. Each scenario is simulated across multiple prediction horizons, the THD, and our proposed AHOD to assess the trade-offs between prediction length, fuel efficiency, and computational load. Then, a detailed evaluation is performed under dense traffic and varying signal cycle durations.

3.1 Scenario 1: Short Red Signal

In the first scenario, the effectiveness of the AHOD is evaluated under a short red signal duration, allowing the vehicle to cross the intersection without a complete stop. The simulation begins with the host vehicle positioned 300 m upstream of the intersection, with a traffic signal cycle consisting of a 15 s green phase followed by a 15 s red phase. Different prediction horizon lengths ($N = 10, 20, 30, 40$) are compared to examine the influence of horizon growth on vehicle trajectory prediction, fuel consumption, and computational performance. The fuel consumption model is derived using the instantaneous velocity and acceleration-dependent polynomial model proposed by Kamal et al. [30], and computation time is measured as the average duration per control iteration.

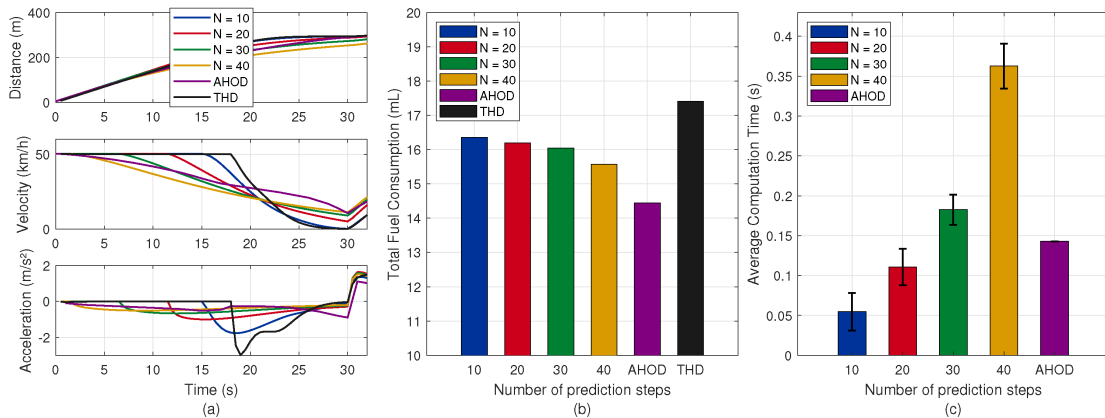


Figure 3. (a) Vehicle distance, velocity, and acceleration trajectories; (b) total fuel consumption; and (c) average computation time for Scenario 1 (short red signal: 15 s green, 15 s red) with varying Model Predictive Control horizons ($N = 10, 20, 30, 40$), traditional human driving model, and adaptive-horizon optimal driving

Figure 3 illustrates the performance results for the short red signal scenario under different horizon lengths ($N = 10, 20, 30, 40$) and the THD compared to the proposed method. With longer prediction horizons, the vehicle

demonstrates earlier anticipation of the red signal, thus minimizing aggressive braking and acceleration (Figure 3a). The THD produces the highest deceleration rate -2.98 m/s^2 , whereas the MPC with $N = 10$ has a maximum deceleration of approximately -1.77 m/s^2 , which gradually decreases to approximately -0.5 m/s^2 as the prediction horizon increases. The AHOD demonstrates anticipatory behavior similar to the MPC, maintaining a maximum deceleration rate of -0.87 m/s^2 . As shown in Figure 3b, total fuel consumption decreases consistently with longer prediction horizons, achieving up to 15.8% improvement relative to THD and 8.9% improvement compared to short-horizon MPC. However, Figure 3c indicates that the average MPC computation time per control step increases with horizon length. On average, computation time scales approximately linearly with N , rising from 0.08 s at $N = 10$ to 0.36 s at $N = 40$. On the other hand, the THD operates with an average execution time of approximately 0.02 s per iteration, and the AHOD takes around 0.14 s for a single optimization run. The proposed AHOD optimizes trajectories comparable to MPC while covering the entire traffic signal cycle length and maintaining lower computational demand.

3.2 Scenario 2: Long Red Signal

The second scenario represents a longer red phase duration. The trajectories in Figure 4a show that the long-horizon MPC effectively anticipates the red phase, resulting in smooth deceleration, while short-horizon MPC and THD exhibit a complete stop. Fuel consumption results (as shown in Figure 4b) demonstrate a benefit of longer prediction horizons, particularly in managing stop-and-go cycles efficiently. The AHOD achieves the lowest total fuel consumption, reducing energy usage by approximately 17.6% compared to THD and 13.7% compared to short horizon MPC. As expected, fuel consumption decreased with increasing prediction horizon, reflecting anticipation of upcoming signal changes. On the other hand, the computational cost trend (Figure 4c) increases with horizon length but remains below 0.4 s per iteration on average. In conventional MPC, each additional prediction step directly adds one decision variable and multiple inequality constraints to the optimization problem solved at every control iteration, causing solver time to increase exponentially. The AHOD avoids such issues because the upper controller maintains a fixed $N = 20$ decision variables regardless of cycle length by extending temporal coverage through non-uniform time steps rather than increasing node count, keeping problem dimension constant. Additionally, the upper controller solves this fixed-size problem only once per signal cycle entry rather than at every control step, while the lower controller handles only a short-horizon optimization at each iteration. The results demonstrate that the optimization problem size increases with the horizon length; however, the AHOD prevents exponential growth in computational complexity and achieves control behavior comparable to that of conventional MPC.

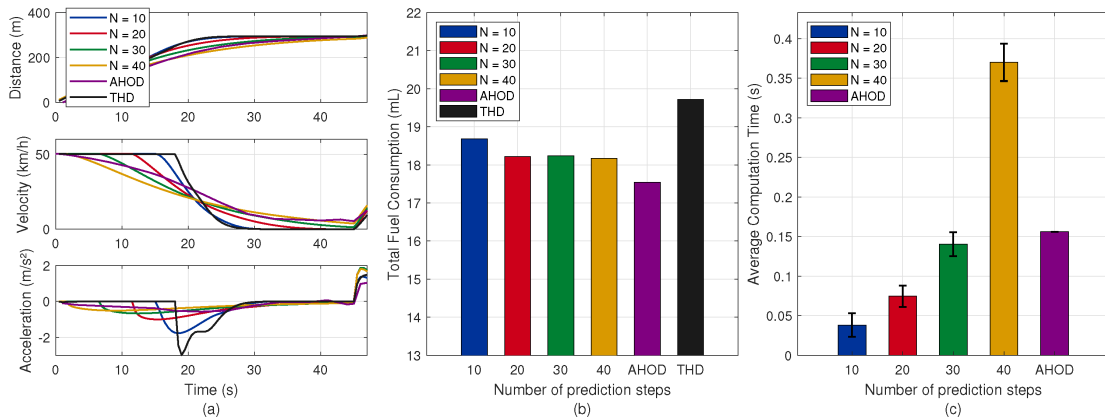


Figure 4. (a) Vehicle distance, velocity, and acceleration trajectories; (b) total fuel consumption; and (c) average computation time for Scenario 2 (long red signal: 15 s green, 30 s red) with varying Model Predictive Control horizons ($N = 10, 20, 30, 40$), traditional human driving model, and adaptive-horizon optimal driving

3.3 Evaluation of Proposed Method in Dense Traffic Conditions

Next, to evaluate the proposed AHOD, we conduct simulations of a single-lane intersection approach in a dense traffic environment and compare it with THD. The signalized intersection is located at 300 m, and a 30 s green (3 s yellow) and 30 s red signal timing configuration is used. Each vehicle control scheme is initialized with identical initial conditions and signal timings to ensure fair comparison. In particular, queue formation emerges naturally from car-following constraints and signal-induced stopping behavior during red phases, while discharge dynamics and start-up lost time arise from vehicle acceleration limits and safe time-headway constraints during green phases. The yellow phase is considered a clearance interval, during which vehicles are prohibited from crossing

the intersection unless they can safely clear the stop line. Spillback effects are captured when upstream vehicles are forced to decelerate due to insufficient downstream spacing. Additionally, a uniform traffic demand of 780 veh/h is considered, with the green signal phase beginning at time 0 seconds. Vehicles approach the intersection at a speed limit of 50 km/h. The comparison of resulting trajectories, velocity, and acceleration is presented in Figure 5.

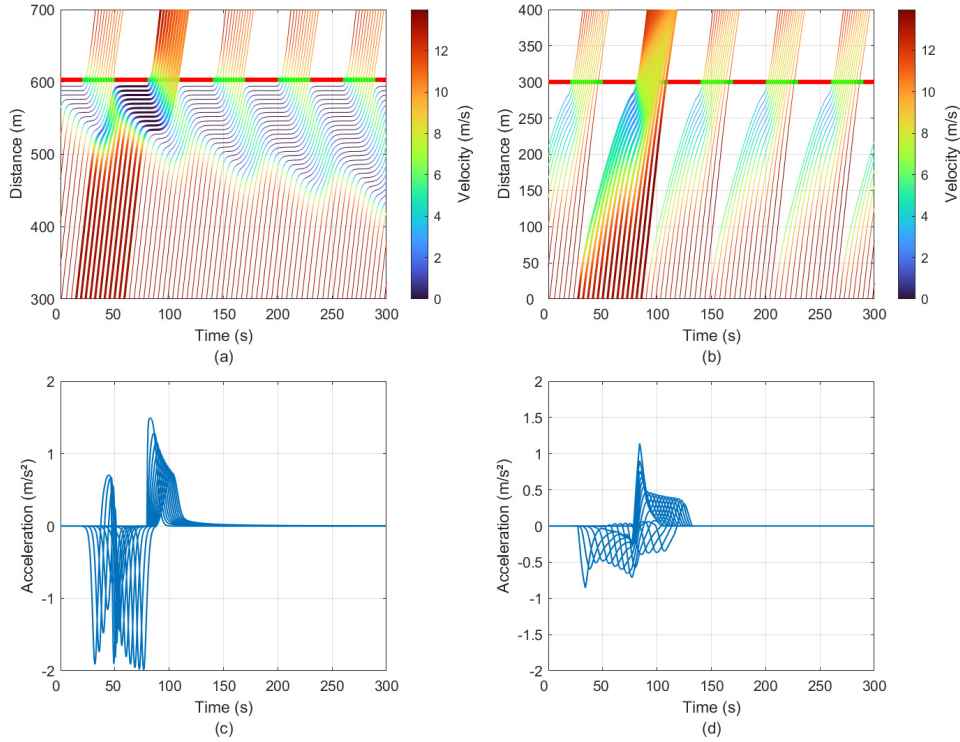


Figure 5. Velocity–distance–time maps for (a) traditional human driving model and (b) adaptive-horizon optimal driving. Acceleration values for successive vehicles during the second green time interval (90–120 s) are shown for (c) THD and (d) AHOD

Note: The color maps represent instantaneous velocity, with warmer colors (red/orange) indicating vehicles traveling at or near the speed limit of 50 km/h and cooler colors (blue) denote deceleration or stopping events.

Figure 5 illustrates trajectory profiles over 300 s of simulation, represented as a velocity heatmap. The color gradient represents instantaneous velocity; warmer colors (red/orange) indicate vehicles traveling at or near the speed limit of 50 km/h, and cooler colors (blue) denote deceleration or stopping events. The AHOD facilitates optimal velocity trajectories that synchronize vehicle arrivals with green signal intervals (Figure 5b). These trajectory profiles are continuous with gradual deceleration and acceleration patterns in anticipation of signal transitions. In contrast, THD-driven vehicles exhibit the conventional stop-and-go behavior associated with human driving, performing full stops during red phases and rapid acceleration at the onset of green. Consequently, THD creates queue formations leading to spill-back congestion, as reflected by the growth of the blue region in the velocity–distance–time heatmap (Figure 5a). The control input (acceleration/deceleration) profiles show the comparison of characteristics of the control. The AHOD generates gradual acceleration inputs that reduce aggressive maneuvers and maintain traffic flow efficiency. In contrast, THD responses exhibit aggressive acceleration and deceleration rates associated with reactive control behavior, leading to increased fuel consumption and traffic delays.

3.4 Evaluation of Proposed Method in Different Traffic Light Durations

Furthermore, we evaluate the adaptability of the proposed method by analyzing vehicle performance under varying traffic-phase durations, specifically 30, 40, and 60 seconds evenly divided between green and red phases. The trajectories in Figure 6 visualize velocity variations along distance and time. Specifically, under the THD control strategy, distinct stop zones can be observed near the intersection during each red phase, characterized by sharp transitions from cruising velocity to idling. These discontinuities in the heatmap correspond to repeated acceleration and deceleration cycles, reflecting inefficient stop-and-go dynamics. In contrast, the AHOD-driven trajectories exhibit smooth and continuous velocity evolution, with gradual color transitions and the absence of prolonged stationary periods, thereby increasing effective green time by reducing start-up lost time.

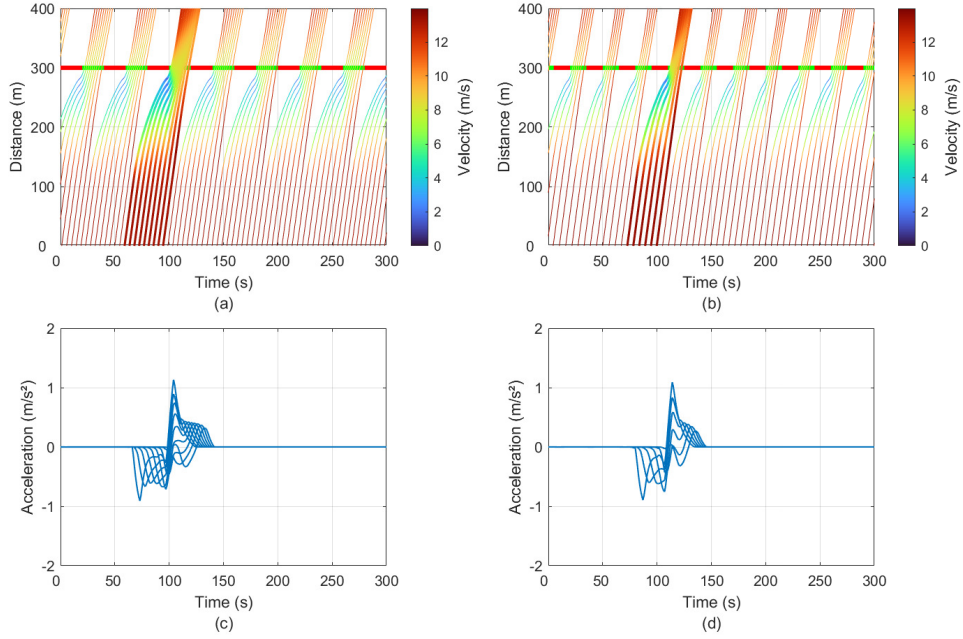


Figure 6. Velocity–distance–time maps for different cycle lengths (a) 40 s and (b) 30 s. Acceleration values for successive cars for (c) the third and (d) fourth green time intervals, respectively

Note: The color maps represent instantaneous velocity, with warmer colors (red/orange) indicating vehicles traveling at or near the speed limit of 50 km/h and cooler colors (blue) denote deceleration or stopping events.

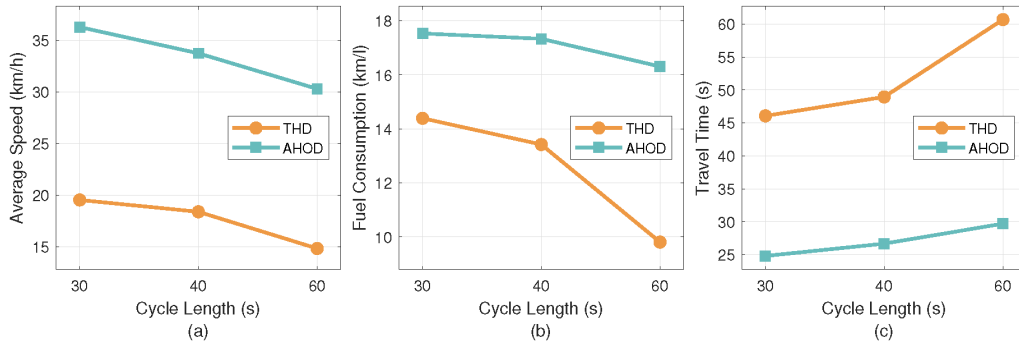


Figure 7. (a) Average speed, (b) total fuel consumption rate, and (c) travel time comparison between adaptive-horizon optimal driving and traditional human driving model for different cycle lengths

Figure 7a presents the average speed comparison for the three signal cycle scenarios. The results consistently demonstrate the advantage of AHOD in all cases. For the 60 s cycle, the AHOD-driven vehicles achieved an average speed of 36.2 km/h compared to 19.5 km/h for THD. This performance advantage results from AHOD's ability to eliminate full stops and maintain continuous motion through anticipatory speed regulation. As cycle length decreases, both methods experience lower average speeds because shorter green intervals limit intersection throughput. However, the performance is significantly better for AHOD. At the 40 s cycle, the average speed is 33.7 km/h for AHOD versus 18.4 km/h for THD, while at the 30 s cycle, the averages are 30.3 km/h and 14.8 km/h, respectively. The relatively horizontal slope of the AHOD curve (speed versus cycle length) indicates that predictive optimization adapts more effectively to varying signal timings, yielding smoother speed transitions.

Figure 7b compares the fuel consumption rate (km/L) across all cycle durations. The AHOD consistently demonstrates significant fuel savings relative to THD. AHOD achieves a fuel economy equivalent to 17.5 km/L, compared to 14.39 km/L for THD at 60 s cycle, corresponding to a 21.6% reduction in consumption. The AHOD strategy reduces fuel consumption associated with abrupt braking and re-acceleration by performing predictive trajectory adjustments and minimizing queue formation during signal phase transitions. Figure 7c depicts the travel time required to cover a 300 m intersection approach segment under both control schemes. The AHOD consistently achieves shorter travel times, reflecting its ability to maintain optimal trajectory and avoid full stops. Importantly,

while THD travel times increase as cycle lengths decrease, AHOD travel times remain relatively stable, which shows that the AHOD can minimize the growth of start-up lost time despite frequent phase changes.

3.5 Discussion

The results highlight a fundamental trade-off between prediction horizon length, computational complexity, and control performance. Longer horizons allow the MPC to capture full signal cycles and anticipate future phase transitions, improving fuel efficiency and trajectory prediction. However, excessive horizon extension provides performance efficiency with increasing solver time and computational load. The adaptive discretization strategy mitigates this issue by coarsening the time resolution for distant predictions, while maintaining accuracy in the near term where control precision is critical. Across all scenarios, the conventional MPC maintained solver times within the 0.4 s limit. However, similar anticipatory behavior can be achieved without performing long-horizon optimization at every control step. These findings suggest that the proposed method provides an effective trade-off between performance and computational cost. Moreover, the AHOD provides a practical foundation for integrating MPC-based decision-making into trajectory planning without excessive computational load.

The generalizability of the proposed AHOD framework is supported by its parametric MPC formulation. The state and input constraints can be adjusted to capture the physical limitations and operational characteristics of different vehicle classes, such as light-duty vehicles, heavy-duty trucks, buses, and electric vehicles, as well as varying driving styles ranging from aggressive to conservative. Due to the common longitudinal kinematic dynamics and shared physical constraints, the relative performance improvements are expected to remain consistent across heterogeneous vehicle dynamics.

While the proposed framework demonstrates consistent improvements across the tested scenarios, it is important to acknowledge the boundary conditions under which performance may degrade. First, for extremely short signal cycles, the available anticipation horizon may be insufficient for the upper controller to generate meaningful long-horizon trajectories, thereby reducing the fuel efficiency advantage over conventional MPC. Second, a sudden unexpected deceleration or stop by a preceding vehicle may render the upper controller's reference trajectory locally infeasible, requiring the lower controller to temporarily override the reference to avoid collision. In such cases, the hard safety constraints in Eq. (11) guarantee collision-free behavior, and a fuel-optimal trajectory may not be fully recovered until the next re-planning event.

4 Conclusion

This study proposes an AHOD framework that reduces computational burden and improves traffic performance through long-horizon trajectory planning. The proposed framework comprises two controllers: an upper controller that generates long-horizon optimal trajectories using an adaptive horizon, and a lower controller that executes real-time safe car-following. The adaptive discretization employs finer time steps in the near term to capture transient dynamics accurately, and coarser steps in the distant horizon to represent longer prediction intervals without computational growth. Simulation results showed that longer prediction horizons enable smoother and more energy-efficient trajectories by allowing vehicles to anticipate upcoming signal phases. However, this improvement comes at the cost of increased problem dimensionality and solver time. By periodically updating the optimal profile rather than solving the MPC problem at every step, the AHOD maintains prediction fidelity without increasing computational demands. Across all cases, the AHOD fuel savings are comparable to long-horizon MPC while operating at significantly lower computational cost. Additionally, the AHOD enhances overall traffic performance, resulting in higher average speeds, reduced travel times, and lower fuel consumption across a range of signal cycle lengths. These findings establish a foundation for integrating long-horizon prediction into MPC-based trajectory planning frameworks.

While the proposed framework provides promising improvements in energy efficiency, mobility, and trajectory planning, several limitations remain. The current framework does not consider V2V communication uncertainties or delays, which may affect real-time performance in practice. Future research will extend the proposed framework to multi-lane and multi-intersection networks, integrating lane-changing behaviors, queue dynamics, and cooperative vehicle interactions. Incorporating communication delays will be essential for real-world applicability.

Author Contributions

Conceptualization, M.A., A.S.M.B., and M.A.S.K.; methodology, M.A., A.S.M.B., and M.A.S.K.; software, M.A.; formal analysis, M.A.; investigation, M.A. and M.A.S.K.; writing-original draft preparation, M.A.; writing-review and editing, M.A., A.S.M.B., M.A.S.K., and K.Y.; visualization, M.A.; supervision, A.S.M.B., M.A.S.K., and K.Y.; project administration, M.A.S.K. and K.Y.; funding acquisition, M.A.S.K. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] T. Jayson, A. S. M. Bakibillah, C. P. Tan, M. A. S. Kamal, V. Monn, and J. I. Imura, "Electric vehicle eco-driving strategy at signalized intersections based on optimal energy consumption," *J. Environ. Manage.*, vol. 368, p. 122245, 2024. <https://doi.org/10.1016/j.jenvman.2024.122245>
- [2] Texas A&M Transportation Institute, "2025 urban mobility report," Texas A&M Transportation Institute, College Station, TX, USA, Tech. Rep., 2025. <https://mobility.tamu.edu/umr/report>
- [3] S. Alshayeb, A. Stevanovic, and J. R. Effinger, "Investigating impacts of various operational conditions on fuel consumption and stop penalty at signalized intersections," *Int. J. Transp. Sci. Technol.*, vol. 11, no. 4, pp. 690–710, 2022. <https://doi.org/10.1016/j.ijst.2021.09.005>
- [4] W. Yue, C. Li, Y. Chen, P. Duan, and G. Mao, "What is the root cause of congestion in urban traffic networks: Road infrastructure or signal control?" *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 8662–8679, 2021. <https://doi.org/10.1109/TITS.2021.3085021>
- [5] Z. Zhang, H. Liu, M. Lei, X. Yan, M. Wang, and J. Hu, "Review on the impacts of cooperative automated driving on transportation and environment," *Transp. Res. Part D Transp. Environ.*, vol. 115, p. 103607, 2023. <https://doi.org/10.1016/j.trd.2023.103607>
- [6] A. Abdelhalim and M. Abbas, "A real-time safety-based optimal velocity model," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 165–175, 2022. <https://doi.org/10.1109/OJITS.2022.3147744>
- [7] M. A. S. Kamal, S. Taguchi, and T. Yoshimura, "Intersection vehicle cooperative eco-driving in the context of partially connected vehicle environment," in *2015 IEEE 18th International Conference on Intelligent Transportation Systems*. Gran Canaria, Spain, 2015, pp. 1261–1266. <https://doi.org/10.1109/ITSC.2015.207>
- [8] H. Jiang, J. Hu, S. An, M. Wang, and B. B. Park, "Eco approaching at an isolated signalized intersection under partially connected and automated vehicles environment," *Transp. Res. Part C Emerg. Technol.*, vol. 79, pp. 290–307, 2017. <https://doi.org/10.1016/j.trc.2017.04.001>
- [9] A. S. M. Bakibillah, M. Hasan, M. M. Rahman, and M. A. S. Kamal, "Predictive car-following scheme for improving traffic flows on urban road networks," *Control Theory Technol.*, vol. 17, no. 4, pp. 325–334, 2019. <https://doi.org/10.1007/s11768-019-9144-z>
- [10] P. Sun, D. Nam, R. Jayakrishnan, and W. Jin, "An eco-driving algorithm based on vehicle to infrastructure (V2i) communications for signalized intersections," *Transp. Res. Part C Emerg. Technol.*, vol. 144, p. 103876, 2022. <https://doi.org/10.1016/j.trc.2022.103876>
- [11] M. Pourmehrab, L. Eleftheriadou, S. Ranka, and M. Martin-Gasulla, "Optimizing signalized intersections performance under conventional and automated vehicles traffic," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 7, pp. 2864–2873, 2020. <https://doi.org/10.1109/TITS.2019.2921025>
- [12] M. Chen, B. Li, Y. Bian, W. Zhuang, S. Evangelou, X. Pan, and B. Chen, "Intersection signal-vehicle coupled coordination with mixed autonomy vehicles," *IEEE Trans. Transp. Electrif.*, vol. 11, no. 1, pp. 668–681, 2025. <https://doi.org/10.1109/TTE.2024.3394595>
- [13] Y. Li, W. Zhang, S. Zhang, Y. Pan, B. Zhou, S. Jiao, and J. Wang, "An improved eco-driving strategy for mixed platoons of autonomous and human-driven vehicles," *Phys. A: Stat. Mech. Appl.*, vol. 641, p. 129733, 2024. <https://doi.org/10.1016/j.physa.2024.129733>
- [14] K. Lu, Y. Chen, Y. Tong, J. Zhang, Y. Luo, and J. Wang, "Eco-driving control for CAVs at signalized intersections: Adapting to traffic uncertainties," *Transp. Res. Part D Transp. Environ.*, vol. 132, p. 104270, 2024. <https://doi.org/10.1016/j.trd.2024.104270>
- [15] X. Liu, J. Ou, D. Yan, Y. Zhang, and G. Deng, "Path tracking control of automated vehicles based on adaptive MPC in variable scenarios," *IET Intell. Transp. Syst.*, vol. 18, no. 6, pp. 1031–1044, 2024. <https://doi.org/10.1049/itr2.12484>
- [16] Z. Chen, J. Lai, P. Li, O. I. Awad, and Y. Zhu, "Prediction horizon-varying model predictive control (MPC) for autonomous vehicle control," *Electronics*, vol. 13, no. 8, 2024. <https://doi.org/10.3390/electronics13081442>

- [17] A. J. Krener, “Adaptive horizon model predictive control,” *IFAC-PapersOnLine*, vol. 51, no. 13, pp. 31–36, 2018. <https://doi.org/10.1016/j.ifacol.2018.07.250>
- [18] Z. Sun, L. Dai, K. Liu, D. V. Dimarogonas, and Y. Xia, “Robust self-triggered MPC with adaptive prediction horizon for perturbed nonlinear systems,” *IEEE Trans. Autom. Control*, vol. 64, no. 11, pp. 4780–4787, 2019. <https://doi.org/10.1109/TAC.2019.2905223>
- [19] C. Zhu, K. Wang, Y. Wang, and S. Li, “Vehicle trajectory adaptive tracking control based on variable prediction horizon,” *Electronics*, vol. 14, no. 9, 2025. <https://doi.org/10.3390/electronics140917699>
- [20] R. C. Shekhar and C. Manzie, “Optimal move blocking strategies for model predictive control,” *Automatica*, vol. 61, pp. 27–34, 2015. <https://doi.org/10.1016/j.automatica.2015.07.030>
- [21] S. H. Son, T. H. Oh, J. W. Kim, and J. M. Lee, “Move blocked model predictive control with improved optimality using semi-explicit approach for applying time-varying blocking structure,” *J. Process Control*, vol. 92, pp. 50–61, 2020. <https://doi.org/10.1016/j.jprocont.2020.04.002>
- [22] D. Garg, M. Patterson, W. W. Hager, A. V. Rao, D. A. Benson, and G. T. Huntington, “A unified framework for the numerical solution of optimal control problems using pseudospectral methods,” *Automatica*, vol. 46, no. 11, pp. 1843–1851, 2010. <https://doi.org/10.1016/j.automatica.2010.06.048>
- [23] A. Makarow, M. Keller, C. Rösmann, and T. Bertram, “Model predictive trajectory set control with adaptive input domain discretization,” in *2018 Annual American Control Conference (ACC)*. Milwaukee, WI, USA, 2018, pp. 3159–3164. <https://doi.org/10.23919/ACC.2018.8431313>
- [24] S. A. Nugroho, V. P. Chellapandi, and H. Borhan, “Predictive eco-driving and eco-charging for connected battery electric vehicles,” *IFAC-PapersOnLine*, vol. 59, no. 30, pp. 509–514, 2025. <https://doi.org/10.1016/j.ifacol.2025.12.288>
- [25] D. Shen, J. Han, D. Karbowski, and A. Rousseau, “Data-driven design of model predictive control for powertrain-aware eco-driving considering nonlinearities using Koopman analysis,” *IFAC-PapersOnLine*, vol. 55, no. 24, pp. 117–122, 2022. <https://doi.org/10.1016/j.ifacol.2022.10.271>
- [26] M. A. S. Kamal, J. I. Imura, T. Hayakawa, A. Ohata, and K. Aihara, “A vehicle-intersection coordination scheme for smooth flows of traffic without using traffic lights,” *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 3, pp. 1136–1147, 2015. <https://doi.org/10.1109/TITS.2014.2354380>
- [27] S. Victor, J. B. Recheur, P. Melchior, and P. Lanusse, “Optimal trajectory planning and robust tracking using vehicle model inversion,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4556–4569, 2022. <https://doi.org/10.1109/TITS.2020.3045917>
- [28] M. Treiber, A. Hennecke, and D. Helbing, “Congested traffic states in empirical observations and microscopic simulations,” *Phys. Rev. E*, vol. 62, no. 2, pp. 1805–1824, 2000. <https://doi.org/10.1103/PhysRevE.62.1805>
- [29] A. S. M. Bakibillah, M. A. S. Kamal, C. P. Tan, T. Hayakawa, and J. I. Imura, “Event-driven stochastic eco-driving strategy at signalized intersections from self-driving data,” *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8557–8569, 2019. <https://doi.org/10.1109/TVT.2019.2931519>
- [30] M. A. S. Kamal, M. Mukai, J. Murata, and T. Kawabe, “Ecological vehicle control on roads with up-down slopes,” *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 3, pp. 783–794, 2011. <https://doi.org/10.1109/TITS.2011.2112648>